

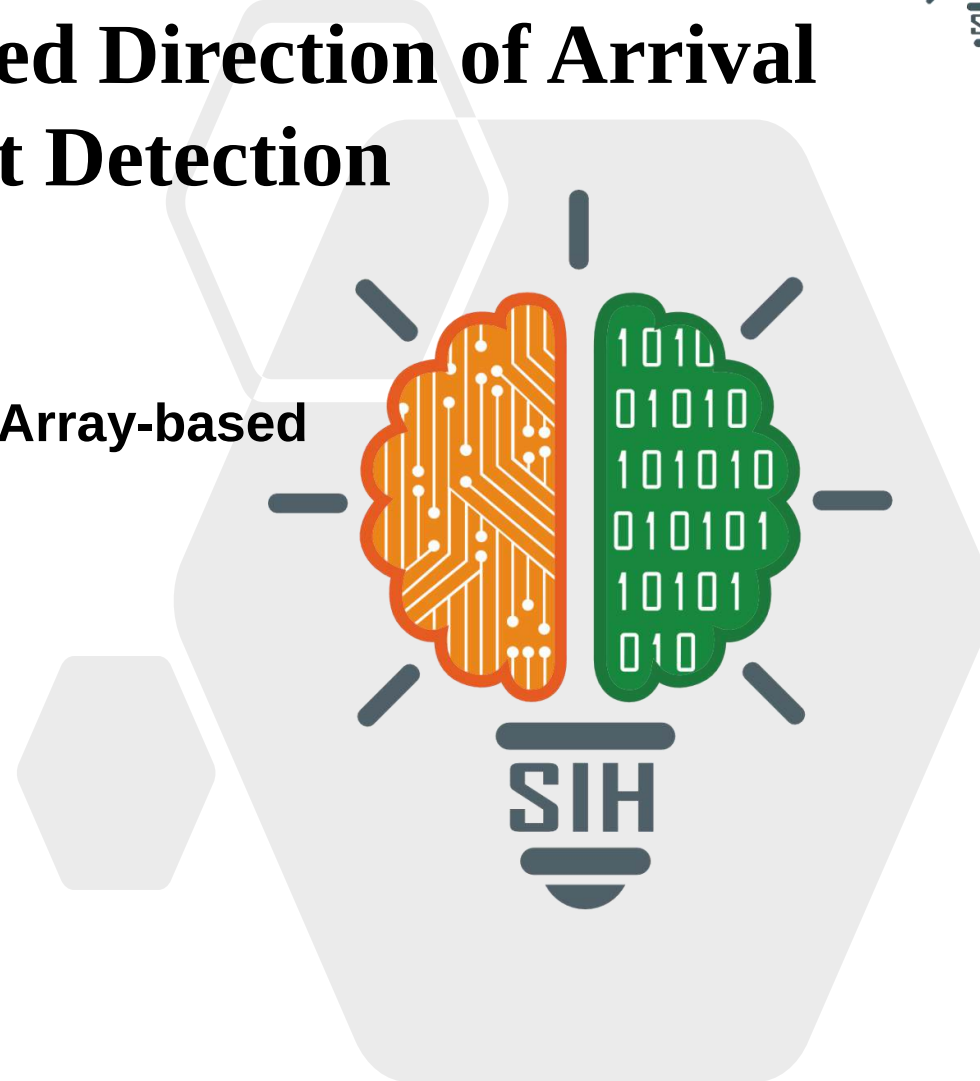
SMART INDIA HACKATHON 2024



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Microphone Array-based Direction of Arrival for Gunshot Detection

- **Problem Statement ID** - SIH1651
- **Problem Statement Title** - Microphone Array-based
Direction of Arrival for Gunshot Detection
- **Theme** - Miscellaneous
- **PS Category** - Software
- **Team ID** – 19974
- **Team Name** – TEAM AURUM



GUNSHOT DETECTION SYSTEM

❖ Proposed Solution:

- Design a Gunshot Detection System using **Zynq 7000 SOC** to detect the Direction of Arrival of the gunshot (Requires both hardware and software design)
- 6 Omni-directional microphones placed in a circular array record the sound signals, which are passed through an ADC then feed them to **Elliptical bandpass filter**
- After applying band-pass filters on the signals, algorithms like **MUSIC (Multiple signal Classification)** and **GCC(Generalized Cross-correlation)** or **Advanced AI&ML algorithms** can be used to deduce the direction from where the bullet has been fired, which is then indicated via a graphical display or a User interface(UI).
- **Edge computing** enables **real-time analytics** on sound data, **low-latency processing** of audio data near the source of the gunshot sound, such as the microphone which is critical for quickly detecting gunshots. By processing data at the edge, you ensure rapid detection and response, which is crucial for security, and safety applications. Using **pre-trained AI/ML models** on edge devices, gunshot detection algorithms can continuously monitor sound patterns and classify sounds

❖ Future Scope:

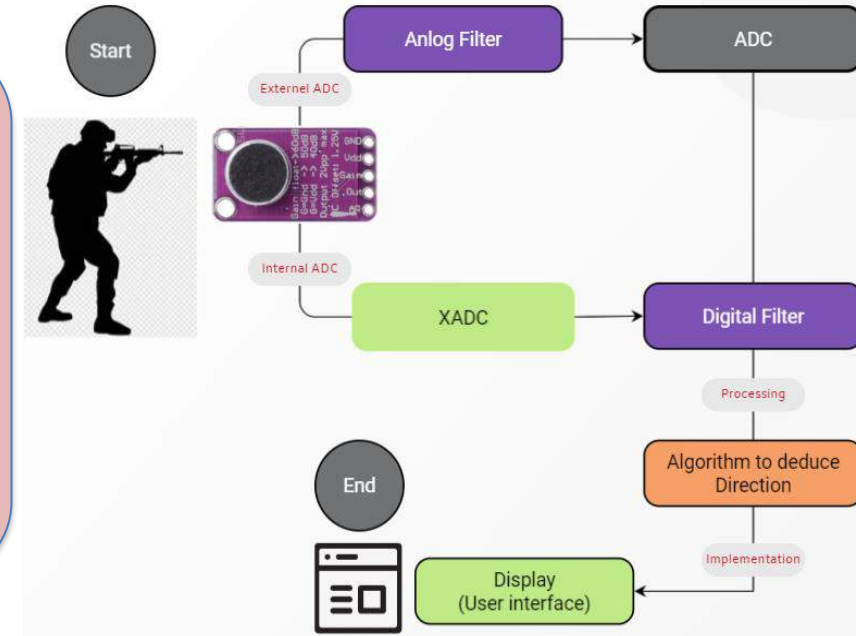
- Using Advanced AI and ML algorithms and **Zynq Ultra Scale+ MPSoC** to achieve better accuracy and efficiency
- We will assess the types of firearms that enemies are equipped with.

❖ Problem Resolution:

- Noise Interference and Distortions
- False Positives
- Power Consumption
 - Environment-based tuning
- High Initial Costs
- Arrangement of Microphone Array

❖ Unique Value Proposition:

- Accuracy and Precision
- Real time processing
- Reconfigurability
- Durability
- Customization
- More efficient due to parallel processing



TECHNICAL APPROACH



❖ Technologies being used:

Hardware :

- FPGA: ZedBoard Zynq-7000 ARM/FPGA SoC Development Board
- Microphone : MAX9814 Omnidirectional Electret Microphone with Amplifier

Software :

- Vitis and Vivado
- MATLAB(Vitis Model Composer)

Languages:

- Verilog HDL and Embedded C

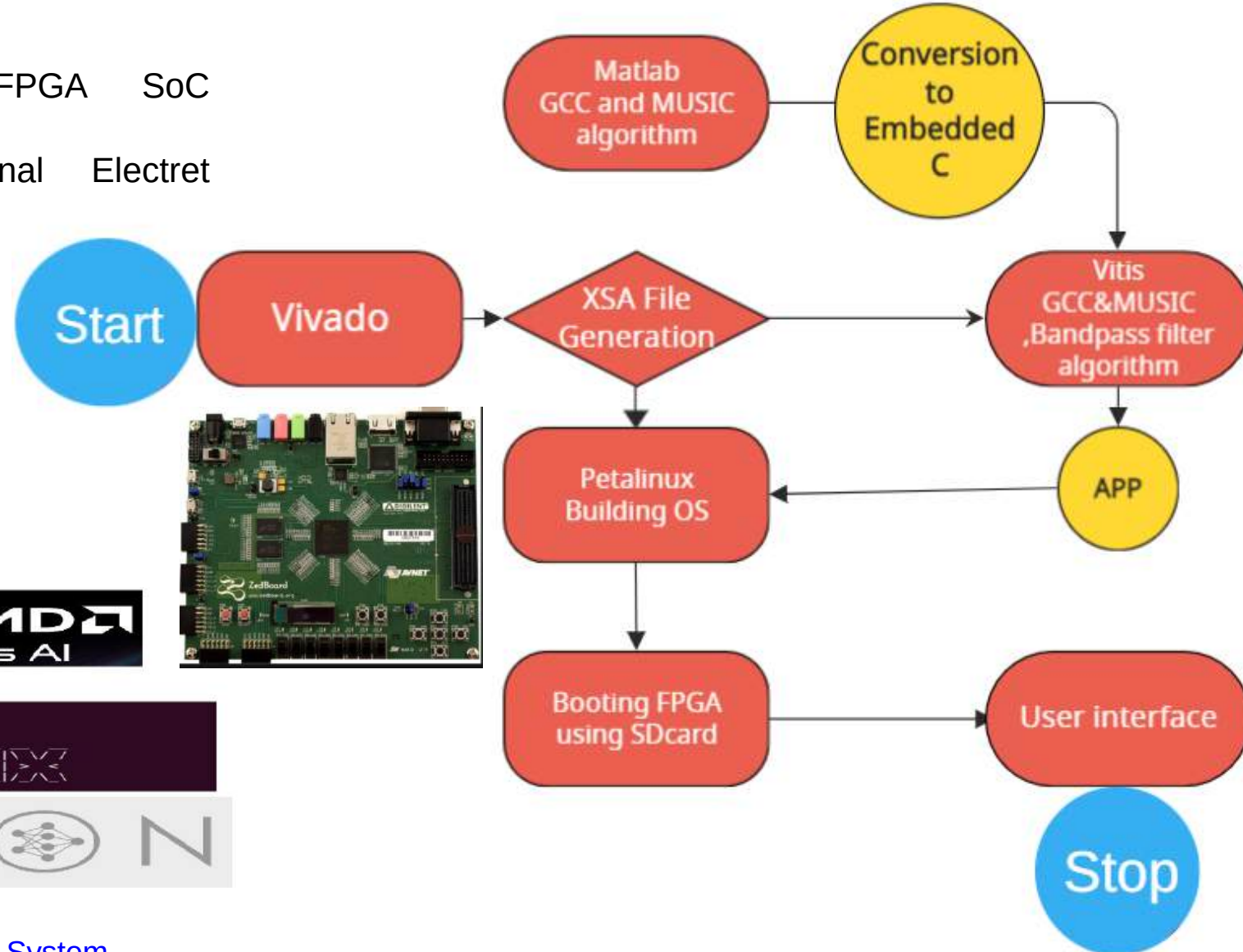
Algorithms:

- GCC(TDOA)(Generalized Cross-Correlation)
- MUSIC(Multiple Signal Classification)



Our progress:

- <https://github.com/tharunreddy1801/Gunshot-Detection-System>



FEASIBILITY AND VIABILITY

FEASIBILITY



- ❖ **Technical:** Microphone arrays with GCC detect gunshot direction; band-pass filters (2-3 kHz) effectively isolate gunshots; real-time processing is achievable with Embedded C or HDL.
- ❖ **Legal Feasibility:** involves ensuring compliance with military standards, intellectual property laws, data privacy regulation, liability concerns, ethical implications, and export control laws

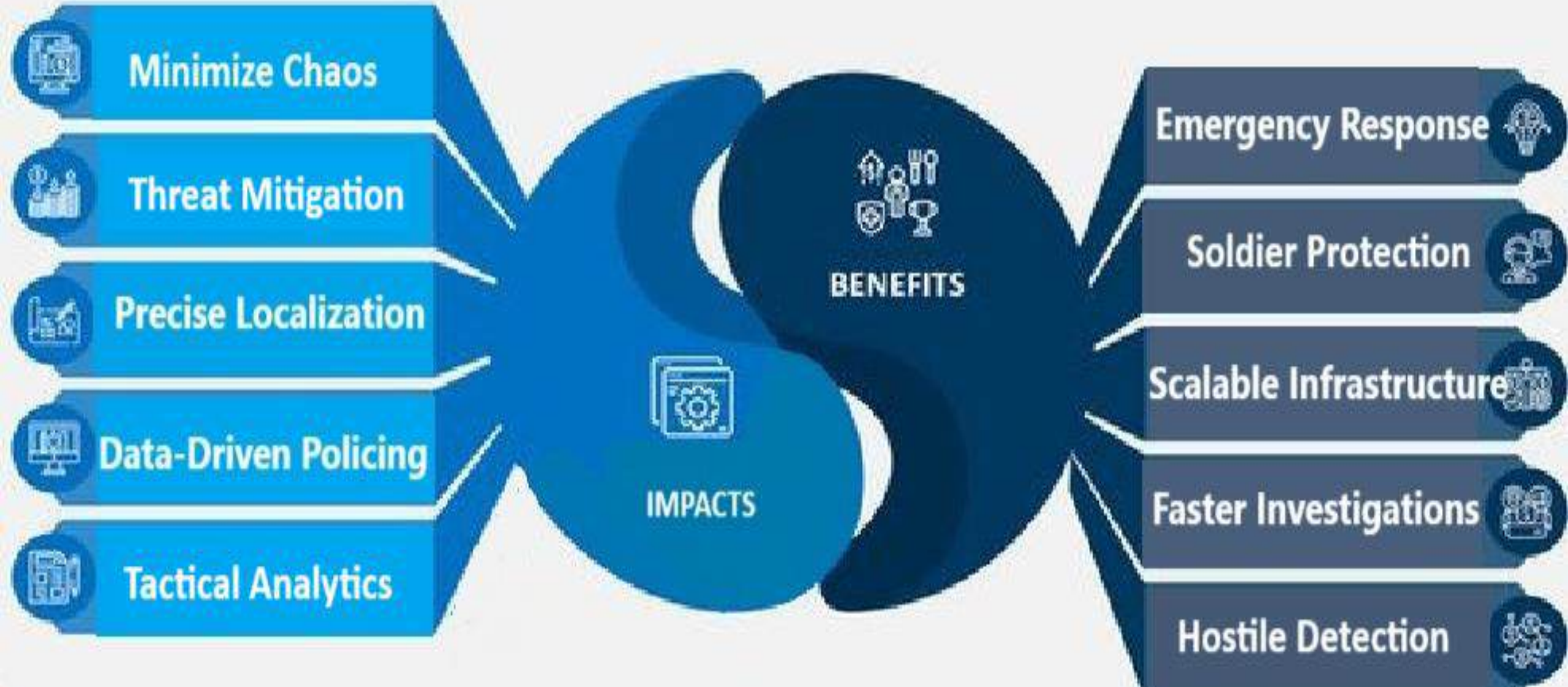
- ❖ **Environmental :** Effective in diverse environments and noise conditions.
- ❖ **Resource:** Utilizes standard microphones and computing resources, minimizing specialized hardware

VIABILITY

- ❖ **Market:** High demand in security; software-based approach offers a competitive, flexible solution.
- ❖ **Scalability:** Easily scalable from small to large deployments; supports future enhancements.
- ❖ **Operational:** Integrates well with existing systems; sustainable with regular updates and minimal maintenance

- ❖ **Financial:** Focusing on affordable software and standard hardware; low ongoing costs.
- ❖ **Usability:** An intuitive user interface can provide rapid detection and localization of gunfire.

IMPACT AND BENEFITS



- **Gunshot Detection System for National Security**(Authors:- Fan Yafeng, Aqeel Mohammed and Marcus Lloyd George)
- **Accurate Time Delay Estimation with the GCC-PHAT**(Author:- Patrick Marmaroli) (<https://www.linkedin.com/pulse/accurate-time-delay-estimation-gcc-phat-patrick-marmaroli-eu7jf/>)
- **MICROPHONE ARRAY BASED SOUND SOURCE DETECTION AND LOCALIZATION WITH BEAMFORMING, GCC-PHAT AND TRIANGULATION USING ARDUINO AND MATLAB**(By Tarun Kadukuntla)
- **Gunshot Airborne Surveillance with Rotary Wing UAV-Embedded Microphone Array**(Authors:-Felipe Goncalves Serrenho , Jose Antonio Apolinario , António Luiz Lopes Ramos 2 and Rigel Procópio Fernandes 3)
- **Gunshot Detection System**(by National protective security authority, U.K.)
- <https://in.mathworks.com/help/phased/ref/gccphat.html>
- <https://in.mathworks.com/help/phased/ug/music-super-resolution-doa-estimation.html>
- <https://www.wired.com/story/ai-gunshot-detection-accuracy-san-jose-nyc/>
- <https://sls.eff.org/technologies/gunshot-detection>
- <https://decibelpro.app/blog/how-far-can-you-hear-a-gunshot/>